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[2]: import pandas as pd
import numpy as np

[3]: df = pd.read_csv('/home/subramanyaprasad/Machine Learning/Data Analysis/Notes/03-Pandas/tips.csv')

[4]: df

[4]:      total_bill  tip    sex  smoker  day  time  size  price_per_person  Payer Name  CC Number  Payment ID
0      16.99    1.01  Female    No  Sun  Dinner    2         8.49  Christy Cunningham  3560325168603410  Sun2959
1      10.34    1.66    Male    No  Sun  Dinner    3         3.45   Douglas Tucker  4478071379779230  Sun4608
2      21.01    3.50    Male    No  Sun  Dinner    3         7.00   Travis Walters  6011812112971322  Sun4458
3      23.68    3.31    Male    No  Sun  Dinner    2        11.84  Nathaniel Harris  4676137647685994  Sun5260
4      24.59    3.61  Female    No  Sun  Dinner    4         6.15    Tonya Carter  4832732618637221  Sun2251
...      ...     ...     ...     ...  ...  ...     ...     ...             ...             ...             ...
239    29.03    5.92    Male    No  Sat  Dinner    3         9.68   Michael Avila  5296068606052842  Sat2657
240    27.18    2.00  Female    Yes  Sat  Dinner    2        13.59  Monica Sanders  3506806155565404  Sat1766
241    22.67    2.00    Male    Yes  Sat  Dinner    2        11.34    Keith Wong  6011891618747196  Sat3880
242    17.82    1.75    Male    No  Sat  Dinner    2         8.91   Dennis Dixon  4375220550950    Sat17
243    18.78    3.00  Female    No  Thur Dinner    2         9.39  Michelle Hardin  3511451626698139  Thur672

244 rows x 11 columns

[5]: df.columns

[5]: Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size',
        'price_per_person', 'Payer Name', 'CC Number', 'Payment ID'],
        dtype='str')
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[12]: df.index

[12]: RangeIndex(start=0, stop=244, step=1)

To get first 5 rows (default):
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```
[13]: df.head()

[13]:      total_bill  tip    sex  smoker  day  time  size  price_per_person  Payer Name  CC Number  Payment ID
0      16.99    1.01  Female    No  Sun  Dinner    2         8.49  Christy Cunningham  3560325168603410  Sun2959
1      10.34    1.66    Male    No  Sun  Dinner    3         3.45   Douglas Tucker  4478071379779230  Sun4608
2      21.01    3.50    Male    No  Sun  Dinner    3         7.00   Travis Walters  6011812112971322  Sun4458
3      23.68    3.31    Male    No  Sun  Dinner    2        11.84  Nathaniel Harris  4676137647685994  Sun5260
4      24.59    3.61  Female    No  Sun  Dinner    4         6.15    Tonya Carter  4832732618637221  Sun2251
```

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To get first n rows:

[14]: df.head(20)
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	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
0	16.99	1.01	Female	No	Sun	Dinner	2	8.49	Christy Cunningham	3560325168603410	Sun2959
1	10.34	1.66	Male	No	Sun	Dinner	3	3.45	Douglas Tucker	4478071379779230	Sun4608
2	21.01	3.50	Male	No	Sun	Dinner	3	7.00	Travis Walters	6011812112971322	Sun4458
3	23.68	3.31	Male	No	Sun	Dinner	2	11.84	Nathaniel Harris	4676137647685994	Sun5260
4	24.59	3.61	Female	No	Sun	Dinner	4	6.15	Tonya Carter	4832732618637221	Sun2251
5	25.29	4.71	Male	No	Sun	Dinner	4	6.32	Erik Smith	213140353657882	Sun9679
6	8.77	2.00	Male	No	Sun	Dinner	2	4.38	Kristopher Johnson	2223727524230344	Sun5985
7	26.88	3.12	Male	No	Sun	Dinner	4	6.72	Robert Buck	3514785077705092	Sun8157
8	15.04	1.96	Male	No	Sun	Dinner	2	7.52	Joseph Mcdonald	3522866365840377	Sun6820
9	14.78	3.23	Male	No	Sun	Dinner	2	7.39	Jerome Abbott	3532124519049786	Sun3775
10	10.27	1.71	Male	No	Sun	Dinner	2	5.14	William Riley	566287581219	Sun2546
11	35.26	5.00	Female	No	Sun	Dinner	4	8.82	Diane Macias	4577817359320969	Sun6686
12	15.42	1.57	Male	No	Sun	Dinner	2	7.71	Chad Harrington	577040572932	Sun1300
13	18.43	3.00	Male	No	Sun	Dinner	4	4.61	Joshua Jones	6011163105616890	Sun2971
14	14.83	3.02	Female	No	Sun	Dinner	2	7.42	Vanessa Jones	30016702287574	Sun3848
15	21.58	3.92	Male	No	Sun	Dinner	2	10.79	Matthew Reilly	180073029785069	Sun1878
16	10.33	1.67	Female	No	Sun	Dinner	3	3.44	Elizabeth Foster	4240025044626033	Sun9715
17	16.29	3.71	Male	No	Sun	Dinner	3	5.43	John Pittman	6521340257218708	Sun2998
18	16.97	3.50	Female	No	Sun	Dinner	3	5.66	Laura Martinez	30422275171379	Sun2789
19	20.65	3.35	Male	No	Sat	Dinner	3	6.88	Timothy Oneal	6568069240986485	Sat9213

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Similarly, to get last ones:

[15]: df.tail()
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	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657
240	27.18	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672

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[16]: df.tail(20)
```

	total_bill	tip	sex	smoker	day	time	size	price_per_person	Payer Name	CC Number	Payment ID
224	13.42	1.58	Male	Yes	Fri	Lunch	2	6.71	Ronald Vaughn DVM	341503466406403	Fri5959
225	16.27	2.50	Female	Yes	Fri	Lunch	2	8.14	Whitney Arnold	3579111947217428	Fri6665
226	10.09	2.00	Female	Yes	Fri	Lunch	2	5.04	Ruth Weiss	5268689490381635	Fri6359
227	20.45	3.00	Male	No	Sat	Dinner	4	5.11	Robert Bradley	213141668145910	Sat4319
228	13.28	2.72	Male	No	Sat	Dinner	2	6.64	Glenn Jones	502061651712	Sat2937
229	22.12	2.88	Female	Yes	Sat	Dinner	2	11.06	Jennifer Russell	4793003293608	Sat3943
230	24.01	2.00	Male	Yes	Sat	Dinner	4	6.00	Michael Osborne	4258682154026	Sat7872
231	15.69	3.00	Male	Yes	Sat	Dinner	3	5.23	Jason Parks	4812333796161	Sat6334
232	11.61	3.39	Male	No	Sat	Dinner	2	5.80	James Taylor	6011482917327995	Sat2124
233	10.77	1.47	Male	No	Sat	Dinner	2	5.38	Paul Novak	6011698897610858	Sat1467
234	15.53	3.00	Male	Yes	Sat	Dinner	2	7.76	Tracy Douglas	4097938155941930	Sat7220
235	10.07	1.25	Male	No	Sat	Dinner	2	5.04	Sean Gonzalez	3534021246117605	Sat4615
236	12.60	1.00	Male	Yes	Sat	Dinner	2	6.30	Matthew Myers	3543676378973965	Sat5032
237	32.83	1.17	Male	Yes	Sat	Dinner	2	16.42	Thomas Brown	4284722681265508	Sat2929
238	35.83	4.67	Female	No	Sat	Dinner	3	11.94	Kimberly Crane	676184013727	Sat9777
239	29.03	5.92	Male	No	Sat	Dinner	3	9.68	Michael Avila	5296068606052842	Sat2657
240	27.18	2.00	Female	Yes	Sat	Dinner	2	13.59	Monica Sanders	3506806155565404	Sat1766
241	22.67	2.00	Male	Yes	Sat	Dinner	2	11.34	Keith Wong	6011891618747196	Sat3880
242	17.82	1.75	Male	No	Sat	Dinner	2	8.91	Dennis Dixon	4375220550950	Sat17
243	18.78	3.00	Female	No	Thur	Dinner	2	9.39	Michelle Hardin	3511451626698139	Thur672

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It calculates some statistical information:

[23]: df.describe()
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	total_bill	tip	size	price_per_person	CC Number
count	244.000000	244.000000	244.000000	244.000000	2.440000e+02
mean	19.785943	2.998279	2.569672	7.888197	2.563496e+15
std	8.902412	1.383638	0.951100	2.914234	2.369340e+15
min	3.070000	1.000000	1.000000	2.880000	6.040679e+10
25%	13.347500	2.000000	2.000000	5.800000	3.040731e+13
50%	17.795000	2.900000	2.000000	7.255000	3.525318e+15
75%	24.127500	3.562500	3.000000	9.390000	4.553675e+15
max	50.810000	10.000000	6.000000	20.270000	6.596454e+15

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Example, 50% of the bills are above 17.79 and 75% of bills are below 24.12

[25]: df.transpose()

[25]:      0      1      2      3      4      5      6
      total_bill      16.99      10.34      21.01      23.68      24.59      25.29      8.77
      tip      1.01      1.66      3.5      3.31      3.61      4.71      2.0
      sex      Female      Male      Male      Male      Female      Male      Male
      smoker      No      No      No      No      No      No      No
      day      Sun      Sun      Sun      Sun      Sun      Sun      Sun
      time      Dinner      Dinner      Dinner      Dinner      Dinner      Dinner      Dinner
      size      2      3      3      2      4      4      2
      price_per_person      8.49      3.45      7.0      11.84      6.15      6.32      4.38
      Payer Name      Christy Cunningham      Douglas Tucker      Travis Walters      Nathaniel Harris      Tonya Carter      Erik Smith      Kristopher Johnson
      CC Number      3560325168603410      4478071379779230      6011812112971322      4676137647685994      4832732618637221      213140353657882      2223727524230344      3514785077705092
      Payment ID      Sun2959      Sun4608      Sun4458      Sun5260      Sun2251      Sun9679      Sun5985

11 rows x 244 columns
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[26]: df.describe().transpose()

[26]:      count      mean      std      min      25%      50%      75%      max
      total_bill      244.0      1.978594e+01      8.902412e+00      3.070000e+00      1.334750e+01      1.779500e+01      2.412750e+01      5.081000e+01
      tip      244.0      2.998279e+00      1.383638e+00      1.000000e+00      2.000000e+00      2.900000e+00      3.562500e+00      1.000000e+01
      size      244.0      2.569672e+00      9.510998e-01      1.000000e+00      2.000000e+00      2.000000e+00      3.000000e+00      6.000000e+00
      price_per_person      244.0      7.888197e+00      2.914234e+00      2.880000e+00      5.800000e+00      7.255000e+00      9.390000e+00      2.027000e+01
      CC Number      244.0      2.563496e+15      2.369340e+15      6.040679e+10      3.040731e+13      3.525318e+15      4.553675e+15      6.596454e+15
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[ ]:
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